Examination of Target Classification for Millimeter-Wave Radar

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0.Abstract

As an alternative to camera-based object identification, which performs poorly in adverse weather, object classification was attempted using millimeter-wave radar and machine learning. Despite lower angular resolution, the radar signals enabled identification of pedestrians, bicycles, and vehicles with 90% accuracy under ideal conditions, confirming the effectiveness of radar signals.

1.introduction

Pedestrian detection using sensors installed on vehicles or roadside infrastructure has been widely studied. However, image-based methods suffer from degraded performance in adverse weather such as heavy fog. Radar, which is more robust to weather and can directly measure distance, provides less angular resolution compared to cameras, resulting in limited information. This limitation can be addressed by temporally accumulating radar-based features, enabling reliable pedestrian detection even in poor weather. We examined an object classification method that accumulates radar data and uses machine learning to distinguish pedestrians, bicycles, and vehicles. This paper presents the classification approach and evaluation results under ideal conditions, demonstrating the potential of radar-based object identification.

2. mm-wave radar

2.1 Principles of Radar

Radar can measure the distance, angle, and speed of a target. Distance is calculated from the round-trip time of the transmitted radio wave. For a target 30 meters away, the round-trip time is only about 0.5 microseconds, which is too short to measure directly. Instead, the FMCW (Frequency-Modulated Continuous Wave) method is used, where the transmission frequency increases linearly over time. The frequency difference between the transmitted and reflected signals, caused by the time delay, allows calculation of the distance. Angle estimation involves changing the direction of transmission and reception. This can be done mechanically by rotating the antenna, or electrically using a phased array antenna. The latter enables MIMO radar systems, which require fewer antenna elements. Speed is determined by detecting phase changes in the received signal. The phase depends on the signal wavelength and the distance to the target, changing 360 degrees for every half-wavelength of distance change. As the target moves, this phase varies over time, and its rate of change reveals the target's speed.

2.2 Millimeter wave radar method

Automotive radar often uses unlicensed millimeter-wave systems classified as specified low-power radio stations. The short wavelength enables compact and lightweight antennas, and the FMCW (Frequency-Modulated Continuous Wave) method, which places minimal load on the transmitter, is widely used for distance measurement. The radar used in this study also follows this approach. For angle measurement, MIMO (Multiple-Input Multiple-Output) systems are adopted, leveraging compact antennas and eliminating the need for mechanical rotation.

2.3 Information that can be obtained

The radar in this study acquires velocity in addition to range and angle. Specifically, it extracts a reflection intensity distribution in a 3D space (defined by range, angle, and velocity axes) from the received signal, as shown in Figure 1 (left). From this distribution, we detect target point clouds by thresholding, assuming that useful targets exhibit high reflection intensity. The detection threshold is calculated for each point in the 3D space based on the surrounding reflection intensities. The coordinates (range, angle, velocity) of the detected points constitute the target information. While the radar initially obtains a target's position in polar coordinates (range and angle), we convert these to Cartesian coordinates as a preparatory step for our identification method. We use Cartesian coordinates because they allow for more intuitive handling of information crucial for identification, such as depth and width.

3. Processing by object label

3.1 Machine learning for object labeling

In the machine learning-based identification process, as shown in Figure 2, target features extracted from radar data by preprocessing (Section 3.2) are fed into an identification rule, which outputs an identification label (pedestrian, bicycle, or vehicle). This identification rule is generated by a learning algorithm, such as Support Vector Machine (SVM) [2], from training data. The training data consists of pairs of features and their corresponding ground truth labels; these features are extracted from pre-recorded radar data for each target type (pedestrian, bicycle, and vehicle) using the same preprocessing method. Generally, a larger amount of training data allows for the construction of a model capable of handling diverse situations.

3.2 Pretreatment

Preprocessing is the signal processing step for extracting features from radar-detected targets that are effective for classification, as illustrated in Fig. 3. After thresholding, multiple reflection points are often detected for a single target. We group these points into clusters based on their spatial proximity in 3D space, which enables the extraction of primary features for each target at every sampling time. Furthermore,

features that span multiple time samples, such as temporal changes in the primary features, are also highly effective for classification. Therefore, we track each target over time by associating clusters from different sampling times that originate from the same target. We then accumulate the primary features extracted from these associated clusters as a time series and use their statistics as secondary features for input to a machine learning model.
As shown in Fig. 4, primary features represent the geometric characteristics of object clusters. The figure compares 3D point clusters of a pedestrian and a vehicle at a specific time. Their spatial distributions differ: for example, the vehicle cluster tends to extend further in the depth direction compared to the pedestrian. Typical speeds also vary significantly between the two. While individual cases differ depending on the situation, such features are considered statistically important for classification.

4. evaluation

To investigate the potential of object classification using millimeter-wave radar, we collected separate datasets for vehicles, bicycles, and pedestrians under ideal conditions without obstacles. First, we verified whether effective features for classification could be extracted from the data. Then, we trained a classification model using those features and evaluated its accuracy.

4.1 Data Collection Environment

As shown in Fig. 5, data were collected in an open area free of buildings and obstacles. The movement directions of the targets included approaching and receding from the radar, as well as crossing from right to left and from left to right. For each direction, multiple scenarios were recorded by varying walking speed and movement conditions.

4.2 Features Used for Evaluation

To verify whether meaningful features can be extracted from the radar data, we arranged the primary features in time series and analyzed their temporal variations. Figure 6 shows the time evolution of the depth-wise distribution of reflection points for a slowly receding vehicle and pedestrian. The lines in the figure represent the maximum, average, and minimum depth positions of reflection points within each cluster. The difference between the maximum and minimum depth values indicates that the vehicle has a wider spread of reflection points than the pedestrian. Additionally, the difference in depth spread between a bicycle and a pedestrian remains relatively stable over time.
Figure 7 shows the temporal change in the velocity distribution of reflection points for a vehicle and a pedestrian. The lines in the figure represent the maximum, mean, and minimum velocities within each cluster. At any given time, the spread of the vehicle's velocity distribution is narrower than that of the pedestrian's. Furthermore, the vehicle's velocity remains stable, whereas the

pedestrian's velocity fluctuates over time.

4.3 Conditions

The accumulation time for primary features was set to 0.5 seconds. The features for our model were the temporal variations in the "spread" and "mean" of reflection point clusters along the range, velocity, and horizontal axes. Although mean velocity is effective for classification, it was excluded from the training data to avoid the risk of the model overfitting to velocity, given the limited number of scenes in our dataset. As shown in Figure 8, radar data from each scene was divided into 0.5-second blocks. Primary features within each block were temporally accumulated to calculate the final features. For the dataset, we extracted a total of 395 blocks: 127 from vehicles, 99 from bicycles, and 169 from pedestrians, all moving individually. These features are classified into one of the three classes (vehicle, bicycle, or pedestrian) based on pre-trained identification rules. These rules were generated using training data composed of feature-label pairs. The classification accuracy is defined as the ratio of correct classifications to the total number of classifications performed every 0.5 seconds. For instance, in the example shown in Figure 8, the accuracy is 80%, with 8 correct classifications out of 10. This indicates that the temporal variations in primary features contain effective information for distinguishing among vehicles, bicycles, and pedestrians.

4.4 Results

The classification results are shown in Table 1. Due to the limited number of data blocks, we used leave-one-out cross-validation (3) for evaluation. The key findings are as follows: Vehicles and Pedestrians: Both were correctly classified with high accuracy (>90%), and misclassifications between these two classes were below 3%. Bicycles: The correct classification rate was below 90%, with over 10% being misclassified as pedestrians. This classification accuracy is not necessarily superior to that of image-based object identification methods. Future work will focus on two areas: Improving Classification Accuracy: This will involve retraining the model with more diverse data, including scenarios with adverse weather, occlusions, and objects with complex movements. We will also explore methods that accumulate information over multiple blocks for decision-making. Exploring Applications: We will investigate specific applications that can leverage the inherent advantages of radar, such as its weather resistance and direct velocity detection capabilities.

5. Conclusion

Using machine learning with features extracted from millimeter-wave radar data, we achieved approximately 90% accuracy in classifying pedestrians, bicycles, and vehicles under low-noise conditions. Features of cars and pedestrians were distinctly different, while bicycles exhibited characteristics of both, indicating that radar data contains useful information for object classification. However, accuracy may degrade in complex scenes involving occlusion or parallel movement, such as bicycles riding alongside pedestrians or pedestrians behind guardrails. To address this, we are collecting data under various conditions and plan to improve accuracy using more advanced learning algorithms. This technology is also applicable beyond traffic scenarios. In environments where privacy concerns prohibit the use of cameras, millimeter-wave radar is a suitable alternative. Such settings present promising opportunities for pedestrian detection using radar-based classification.

まず、それぞれのセクションごとに区切り、1.Google AI Studio Gemini 2.5 Pro Preview 05-08 と 2.ChatGPT 4-o を利用してそれぞれに「この文章から 1、重複表現を減らす。 2、無駄な言葉や言い回しをなくす。 その上で英訳してください。 すべての過程を表示してください」と入力した。出力されたものをさらに原文と英文それぞれを Gemini Pro 2.5 に入れた上で、「上は論文の原文です。以下に翻訳した英文を 2 つ挙げます。技術論文として英訳する場合に良いかどうか比較してください。」と入力し、論理的によいと判断した方を適用した。

以下では 1=Google AI Studio Gemini 2.5 Pro Preview 05-08 と 2=ChatGPT 4-o と表す。

まずまえがき。専門用語と表現について。「歩行者を識別する技術」を 1「Pedestrian identification technology」2「Object classification」の二候補であった。1 の identification は原文の「識別」に最も忠実な訳語であるが、論文のメインは対象物を見つける「検知」に限らず「歩行者・自転車・自動車」といった複数のカテゴリーに分類することにある。そのため、object classification のほうが適切であると判断した。
「耐候性が優れた」
1「excellent weather resistance」2「more robust to weather」
どちらの表現も意味的に正しいが、robust という表現は技術分野において「外部環境の変化に強い」という意味でよく用いられることから、2 を採用した。
「角度方向の分解能が劣る」
1「its angular resolution is inferior to images」2「provides less angular resolution compared to cameras」
Inferior to より、less compared to のほうが比較表現としてスムーズであるから 2 を採用した。

第一段落について、文章全体として、2 は情報は失わずに、より少ない単語で同じ意概念を説明している。技術論文において読者が素早く内容を理解する必要があるので重要である。2 では This can be done という表現が用いられており、明快である。
速度の説明においても its rate of change reveals the target’s speed とあり、現象と結果の関係性を表していて専門的で洗練されている。
原文には FMCW 方式を説明するための具体例が含まれている。翻訳 1 は全て省略していたが、2 は正確に含めており、読者の理解を助けている。
専門用語の選択について、1「MIMO … is also known」2「The latter enables MIMO radar systems…」1 は「知られている」の直訳。2 はフェーズドアレイ方式が MIMO という技術的発展にどうつながるかを能動的に示している。

2章について、翻訳1ではas shown in Figure 1(left)という記述があるが、2は無い。技術論文において図表への言及は重要なので省略してはいけない。さらに、文章全体として、1は全ての内容を一つの段落にまとめているが、2は一行ずつ箇条書きのように分割しており論文としては不適切。
表現について、「距離と角度であるが…直交座標を用いる」1「target positions in terms of range and angle, … utilizes Cartesian coordinates」2「position is initially obtained in polar coordinates (distance and angle), we convert them to Cartesian coordinates」
表現としては「距離と角度」をpolar coordinates（極座標）という専門用語を使って明確にしている。動詞の選択について、1「a transformation … is performed」2「we convert them」能動態であるwe convertのほうが主体が明確である。技術論文では能動態表現がよい傾向がある。そのためその翻訳を適用した。

3.1節について、今回利用した翻訳1はGemini 2.5 Proは「専門用語を的確に用いながら原文の論理的な流れを自然な英語の段落として再構成しており非常に完成度が高い」と評している。2は情報を単純化しすぎで論文として不可欠な要素が欠けていると評していた。具体的には、原文「サポートベクターマシン**(2)**を用いた。」1「Support Vector Machine(SVM)[2]」2「We used a Support Vector Machine(SVM)」とあり、引用を完全に省略している。専門用語の選択においても、原文「物標の識別ラベルの組」翻訳1「pairs of features and their corresponding ground truth labels」翻訳2「feature values and their corresponding labels」とあるが、機械学習の分野で、学習データに付与する「正解ラベル」のことをground truthと呼ぶのが一般的である。原文「識別ルールに入力する」翻訳1「are fed into an identification rule」翻訳2「are input to a classification rule」では、機械学習のモデルやプロセスにデータを「入力する」という意味でfeedという動詞は用いられ、より自然でこなれた表現とのこと。さらに文章全体の構成としても、今回利用した翻訳1のほうが論理的なつながりを重視して構成されているので翻訳1を利用した。

3.2節について、1「… clustering is performed based on the 3D distances… 」（クラスタリングが実施される）2「We group these points into clusters based on their spatial proximity…」（我々は点をクラスタにグループ分けする）。翻訳2のWe groupは研究の主体が何をしたのかを明確に示している。また、翻訳1「… tracking is performed…」（追尾が実施される）翻訳2「…we track each object…」（我々は各物体を追尾する）とあり、we trackのほうが能動的である。しかし文章のフォーマットとして2は短い文章として文が構成されており論文の本文として不適切であった。よって単語の選び方は翻訳2のものを利用し翻訳1のフォーマットで書き直した。
次の段落は、2の翻訳が、「能動態による明確な主体性」「専門用語の的確さ」「表現の簡潔さ」

の点において優れていると評していたのでこちらを利用した。具体的には、1「…data … were individually acquired…」（データが個別に取得された）2「… we collected separate datasets…」（我々は別々のデータセットを収集した）とあり、we collected のほうが誰が研究を行ったのかを明確に示す。現代の論文で好まれるとある。さらに 1「This study verifies…, then generates… and evaluates… 」（この研究は検証し、生成し、評価する）2「First, we verified… Then, we trained… and evaluated…」（最初に我々は検証し、次に我々は学習させ、評価した）とあり、翻訳 2 は「First… ,Then…」という構成と能動態 we…を組み合わせることで、研究ステップが時系列でわかりやすく読者が理解しやすいとあった。専門用語の翻訳も、「物標識別」1「target identification」2「object classification」機械学習の分野において複数のクラス（自動車、自転車、歩行者）に“分類”は classification と呼ぶのが一般的である。「識別率」1「identification rate」2「accuracy」機械学習の性能評価指標としては accuracy（正解率）が一般的で的確な用語である。「データ取得環境」1「environment free of obstacles or structures near the road」（道路の近くに障害物や建造物がない環境）2「open area free of buildings and obstacles」（建物や障害物のないひらけた場所）、open area のほうが日本語の意図を汲み取っている。これらのことから 2 の翻訳を選択した。

4.2 章は、2 の翻訳が「語彙の選択の的確さ」「表現の自然さ」から選択した。原文「…有効な特徴が抽出できるかを検証するため…」1「To verify if effective features… can be extracted…」2「To verify whether meaningful features can be extracted…」effective（効果的な）もよいが、「意味のある、有意義な」特徴量を見つけ出せるかというニュアンスが強いため、meaningful のほうが意図を的確に捉えている。原文「…その時間変化に注目する」1「…their temporal changes are analyzed. 」（時間変化が分析される）2「… we analyzed their temporal variations. 」（我々は時間変化を分析した）、これも能動態 we analyzed のほうが研究主体が明確である。さらに、changes よりも variations のほうが、時系列データがどのように変化していくかというニュアンスを伝えている。原文「…位置分布の時間変化を示している。」1「… the temporal change in the distribution of reflection point positions…」2「…the time evolution of the … distribution…」temporal change は直訳として正しいが、time evolution（時間発展、時間的推移）は信号処理の分野で、ある量が時間とともにどう変化していくかを表すときによく使われ、洗練された表現である。原文「…奥行き方向の位置の最大値と最小値の差を比較すると…」1「A comparison of the spread in the depth-wise distribution… shows that…」2「The difference between the maximum and minimum depth values indicates that…」1 は少し回りくどいが、2 は「最大値と最小値の差が〜を示す」と原文構造に沿っておりわかりやすい。翻訳 1 も原文の意味は伝わるが、技術論文としては翻訳 2 のほうが的確な表現を用いているのでこちらを用いた。

4.3 では、専門用語の選択、簡潔さ、表現の自然さから 1 を利用した。原文「…データ取得の

シーン数が少なく、速度のみで識別できてしまう可能性が高いため…」1「… to avoid the risk of the model overfitting to velocity, given the limited number of scenes…」2「… there was a high possibility that classification could rely solely on velocity」rely solely on velocity は原文に忠実な翻訳であるが、機械学習という分野において、「学習データが少ないために、特定の特徴だけで識別できてしまう状態」を専門用語で overfitting（過学習）と呼ばれる。よって 1 の方がよい。さらに、データ内容の表現も、1「…we extracted a total of 395 blocks: 127 from vehicles, 99 from bicycles, and 169 from pedestrians…」2「…127,99, and 169 blocks are obtained, respectively, resulting in a total of 395 blocks. 」1 のほうがコロンを利用して情報をリスト化しており、読みやすく、英語技術論文で頻繁に使われるスタイルである。識別率の定義という部分も 1「The classification accuracy is defined as the ratio of…」2「The classification accuracy was calculated as the ratio of…」用語の定義を述べる場合、is defined as のほうが、was calculated as よりもフォーマルかつ一般的である。

4.4 章では、結果の章であり、原文の情報を再構成し、箇条書きを用いることにより結果と今後の課題を瞬時に把握できるようになっている。また翻訳 2 は引用番号がなく、論文として不適切である。さらに結果の要約も 1「Both were correctly classified with high accuracy(90%)」2「When features from cars were used as input, over 90% were…」1 は自動車と歩行者の結果をまとめている。2 は原文通りに一つずつ説明しており、正確であるが、要約されておらずわかりにくい。
5 章のあとがきは、どちらの翻訳もよいが、専門用語の選択と表現の洗練度で 1 が優れていたため選択した。原文「自動車と歩行者の特徴は大きく異なっており…」1「The radar signatures of vehicles and pedestrians differ significantly…」2「Features of cars and pedestrians were distinctly different…」特徴の訳として Features は正しい。しかし、レーダーの分野では、物体から返ってくる特有の反射パターンや信号特性を signature と表現することはある。これは「特徴」よりも「その物体固有の識別情報」というニュアンスを含んだ洗練された用語である。原文「外乱が少ない状況」1「under favorable conditions with minimal interference」2「under low-noise conditions」1 の favorable conditions（好ましい条件）と minimal interface（最小限の干渉）は状況をより具体的に描写している。これらのことから 1 を用いた。